

SPECIAL ISSUE ARTICLE

Developing a New I - V Translation Methodology in Accordance With IEC 60891:2021 Correction Procedure 1 and 2

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ABSTRACT

Accurate translation of photovoltaic (PV) module I - V curves under varying operating conditions is essential for evaluating module performance and ensuring reliable energy output predictions. This study proposes a hybrid correction procedure (HCP) for I - V curves, utilizing a four-parameter Gompertz model to integrate the strengths of CP1 and CP2, thereby unifying the two approaches into a cohesive framework. It further evaluates the performance of HCP alongside existing correction methods (CP1 and CP2) across a broad range of irradiance (G) and module temperature (T_m) conditions. CP1 demonstrates the highest accuracy in P_{MAX} correction, with RMSE values consistently below 0.50% for most modules, but fails to generate complete I - V curves when translating from low to high-irradiance levels. CP2, while achieving superior accuracy in V_{OC} correction, struggles with low- R_{sh} modules, leading to distorted I - V curves and reduced accuracy in P_{MAX} and I_{SC} . The proposed HCP method successfully integrates the strengths of CP1 and CP2. It achieves I_{SC} results comparable to CP1, V_{OC} results similar to CP2, and significantly lower RMSE in P_{MAX} compared to CP2, particularly under challenging low- R_{sh} and high- R_s conditions. These findings validate the effectiveness of HCP as a robust alternative correction method for improving the accuracy of I - V curve translations under diverse real-world operating conditions. This work provides a pathway to more accurate PV module performance analysis and highlights the importance of addressing R_{sh} and R_s effects in future correction procedures.

1 | Introduction

By the end of 2024, the global cumulative installed capacity of photovoltaic (PV) systems exceeded 2200 GW, with over 550 GW of new installations added during that year alone [1]. This remarkable growth is driven by the advancements in high-efficiency PV technologies, including heterojunction (HJT) and Tunnel Oxide Passivated Contacts (TOPCon), with the potential integration of perovskite-based PV devices expected to further revolutionize the industry in the future. The current-voltage

characteristics (I - V curves) serve as a fundamental method for assessing the electrical performance of PV devices, providing crucial information for performance assessment. From these I - V curves, key electrical parameters such as short-circuit current (I_{SC}), open-circuit voltage (V_{OC}), maximum power (P_{MAX}), series resistance (R_s), and shunt resistance (R_{sh}) can be extracted for further diagnosis and analytical evaluations.

In the PV industry, the power rating of PV devices is typically determined under standard test conditions (STC), which define

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a device temperature of 25°C and an irradiance of 1000 W/m² with a specified spectrum defined in IEC 60904-3 [2]. However, actual measurement conditions almost always deviate from STC, particularly during outdoor measurements. On-site *I-V* curve measurements necessitate the translation of *I-V* curves obtained under various environmental conditions to a reference condition, typically the STC, to enable consistent comparisons and the extraction of key parameters for fault diagnosis or power loss analysis [3].

To ensure comparability across varying conditions, *I-V* curve correction becomes indispensable. Recognizing the necessity for precision, several correction methods are proposed. The most commonly used methods are those from the IEC 60891 standard. The first edition of IEC 60891, introduced in 1987, proposed an *I-V* curve correction method known as Correction Procedure 1 (CP1) to enhance the accuracy of results. The second edition of IEC 60891, published in 2009, introduced two additional methods: Correction Procedure 2 (CP2) and Correction Procedure 3 (CP3). In the most recent version of IEC 60891, released in 2021, Correction Procedure 4 (CP4), developed by Y. Hishikawa et al. [4], was added, while CP2 was updated [5] based on the work of Monokroussos et al. [6]. CP1, CP2, and CP4 are all based on the single-diode model (SDM), whereas CP3 employs a different approach, utilizing multiple *I-V* curves to perform linear interpolation or extrapolation for correction, as proposed by Hishikawa Y. et al. [7].

The performance of those *I-V* correction methods has been extensively studied and evaluated in numerous research. Duck et al. [8] evaluated the performance of P_{MAX} translation for three CPs outlined in IEC 60891:2009 using outdoor data. Raina et al. [9] optimized determination method of R_s and curve factor (κ), while also assessing the applicability of CP2. Li et al. [10] conducted a comparative analysis of the CP1, CP2, and CP4. Xu et al. [11] performed a comprehensive assessment of all four CPs presented in the IEC 60891:2021, considering a wide range of irradiance (G) and temperature (T_M) conditions. Pó et al. [12], in their study of CP1 and CP2 (version 2009), indicated a systematic bias in the power derived from translated *I-V* curves. Additionally, Dobreava et al. [13] revealed that the mathematical foundation of *I-V* curve translation lies in affine transformations of concave functions defined on convex sets.

Among the correction procedures, both CP1 and CP2 have demonstrated high accuracy across diverse irradiance and temperature ranges. CP2 offers the advantage of consistently generating complete *I-V* curves when translating to higher irradiance levels. However, it is less suitable for PV devices with low R_{sh} and requires more input parameters [11]. In contrast, CP1 requires measurement of *I-V* curves extending adequately into the negative current regime at low-irradiance levels to produce complete *I-V* curves when performing translations to higher irradiance levels [6, 11].

Extensive research has been focused on optimization of CP2 to enhance translation accuracy and expand applicability. Li et al. [14] introduced a method to improve the performance of CP2 (version 2009) under fault conditions. Furthermore, Li et al. proposed an improved *I-V* correction procedure based on the latest

CP2 (version 2021), which improves the accuracy for PV devices with low R_s and low R_{sh} . Xu et al. [11] proposed an approach to increasing the accuracy of CP2 for PV devices with low R_{sh} . However, these studies face limitations in accurately extracting R_{sh} from the original measured *I-V* curves.

This study presents an innovative approach that addresses the limitations of CP1 and CP2 by integrating their strengths using a four-parameter Gompertz curve. In previous research, A. Molina-García et al. [15] employed the Gompertz curve to model the *I-V* characteristics of PV modules as an alternative to traditional single- or double-diode models. The present work utilizes the Gompertz curve as a sigmoid function to effectively unify and enhance the capabilities of CP1 and CP2. The proposed method aims to provide a more robust and universally applicable *I-V* translation method enhancing the accuracy of PV power rating assessments under varying measurement conditions. Validation of the new method is conducted using experimental datasets spanning a range of temperatures and irradiance levels, with particular emphasis on Tunnel Oxide Passivated Contact (TOPCon) solar cell technology. Additionally, this study utilizes a simulated PV module based on Bishop's model [16] to assess the translation accuracy for PV devices exhibiting low R_{sh} and high R_s .

2 | Methodology

2.1 | *I-V* Curve Correction Procedure

As discussed in the previous section, the most widely employed *I-V* curve correction methods are CP1 and CP2, as proposed in IEC 60891. This paper introduces a hybrid correction procedure (HCP), developed based on the principles of CP1 and CP2. All three correction procedures are outlined below and collectively evaluated in the subsequent sections of this study.

2.1.1 | CP1

$$I_2 = I_1 + I_{SC1} \left(\frac{G_2}{G_1} - 1 \right) + \alpha (T_2 - T_1) \quad (1)$$

$$V_2 = V_1 - R_s (I_2 - I_1) - \kappa I_2 (T_2 - T_1) + \beta (T_2 - T_1) \quad (2)$$

where I_1 , V_1 , T_1 , and G_1 are the measured current, voltage, module temperature, and irradiance, respectively; I_2 and V_2 are the corresponding pair of current and voltage values of the corrected *I-V* curve; T_2 and G_2 are the target module temperature and irradiance, respectively; I_{SC1} is the measured short-circuit current; R_s is the internal series resistance; α and β are, respectively, the short-circuit current and open-circuit voltage temperature coefficients (TC) at the target irradiance for correction; and κ is the curve correction factor.

R_s and κ are the predetermined correction parameters, which can be calculated using a group of *I-V* curves at the same temperature and irradiance, respectively.

2.1.2 | CP2

The equations used in CP2 are as follows:

$$I_2 = \frac{G_2}{G_1} I_1 \frac{(1 + a_{\text{rel}}(T_2 - 25))}{(1 + a_{\text{rel}}(T_1 - 25))} \quad (3)$$

$$V_2 = V_1 - R'_{S1}(I_2 - I_1) - \kappa' I_2 (T_2 - T_1) + V_{\text{OC,STC}} \left\{ \beta_{\text{rel}} [f(G_2)(T_2 - 25) - f(G_1)(T_1 - 25)] + \frac{1}{f(G_2)} - \frac{1}{f(G_1)} \right\} \quad (4)$$

$$R'_{S1} = R'_S + \kappa' (T_1 - 25) \quad (5)$$

$$V_{\text{OC,STC}} = \frac{V_{\text{OC}} f(G_1)}{1 + \beta_{\text{rel}}(T_1 - 25) f^2(G_1)} \quad (6)$$

$$f(G) = \frac{V_{\text{OC,STC}}}{V_{\text{OC}}(G)} = B_2 \ln^2 \left(\frac{1000}{G} \right) + B_1 \ln \left(\frac{1000}{G} \right) + 1 \quad (7)$$

where $V_{\text{OC,STC}}$ is the open-circuit voltage at STC; β_{rel} is the relative open-circuit voltage TC; R'_S is the internal series resistance determined at 25°C; R'_{S1} is the internal series resistance at measured temperature T1; it can be calculated from Equation (10); κ' is the TC of the internal series resistance R'_S ; and B_1 and B_2 are the irradiance correction factor for V_{OC} , which accounts for non-linearity of V_{OC} with irradiance.

Similar to CP1, B_1 , B_2 , R'_S , and κ' are the predetermined correction parameters, which can be calculated using a group of I - V curves at the same temperature or same irradiance.

2.1.3 | HCP

A four-parameter Gompertz curve is implemented to integrate the outcomes from CP1 and CP2, thereby harmonizing the two approaches into a unified framework. The Gompertz curve, widely utilized in biological and growth studies, has been suitably adapted for this purpose. The four-parameter Gompertz model, originally introduced by Gibson et al. [17, 18], is expressed using the following notation:

$$Y(V) = A + B \exp(-\exp(k(V - V_k))) \quad (8)$$

In this work, the Gompertz curve is utilized as a sigmoid function, commonly applied in machine learning, to map any numerical value within the range of 0–1. By adjusting the parameters A and B , the four-parameter Gompertz curve smoothly transitions from 0 to 1 along the positive x-axis.

For this study, two distinct Gompertz curves were generated, corresponding to the translated currents produced by CP1 and CP2. These curves were combined by multiplying each curve with its respective translated current, resulting in two sets of current points. The summation of these sets defines the final current points, which represent the new I - V correction procedure. The new I - V curve achieves a smooth transition between CP1 and CP2 by fine-tuning the parameters k and V_k . This

approach enables the new I - V curve to effectively integrate the strengths of CP1 and CP2, allowing the I_{SC} and P_{MAX} regions to align closely with CP1, while the V_{OC} region aligns more closely with CP2. The graph of Gompertz curve is provided in Supporting Information S1: Section A. The two Gompertz curves are expressed as follows:

$$Y1(V) = \exp(-\exp(k(V - V_k))) \quad (9)$$

$$Y2(V) = -1 + \exp(-\exp(k(V - V_k))) \quad (10)$$

where V is the voltage point of the original measured I - V curve; V_k determines the curve at which voltage starts increasing or decreasing; and k determines the increasing or decreasing rate; a larger value indicates higher rate. The recommended value is from 0.3 to 0.8. In this paper, k is always set to 0.5.

The value of V_i is applied by the following formula in this paper with:

$$V_k = V_{\text{MPP1}} \times i \quad (11)$$

where V_{MPP1} is the maximum power point voltage of the measured I - V curve, and i is the coefficient that depends on the difference between the P_{MAX} produced CP1 and CP2. The value of i is listed in Table 1, which is semiempirical. The determination of i is detailed in Supporting Information S1: Section B.

The HCP is defined in the following formulas:

$$I_f = Y1 \times I_{\text{CP1}} + Y2 \times I_{\text{CP2}} \quad (12)$$

$$V_f = V_{\text{CP2}} \quad (13)$$

where I_f and V_f are the final translated current and voltage dataset, respectively; I_{CP1} and I_{CP2} are the translated current dataset produced by CP1 and CP2, respectively; V_{CP2} is the translated voltage dataset produced by CP2.

2.2 | Test Samples

To comprehensively evaluate the performance of three correction methods, four different conditions are studied: (i) healthy,

TABLE 1 | Parameters setting for the simulated modules.

Sample	$R_{\text{sh}} (\Omega)$	$R_s (\Omega)$
Healthy	1500	0.1
LRsh#1	150	0.1
LRsh#2	100	0.1
LRsh#3	50	0.1
HRs#1	1500	0.5
HRs#2	1500	0.75
HRs#2	1500	1
LH#1	100	0.75
LH#2	50	1

(ii) high R_s , (iii) low R_{sh} , and (iv) low R_{sh} and high R_s . To eliminate the influence of all other sources of uncertainty, in particular the measurement errors related to irradiance and temperature, these modules are generated through simulations.

To generate different condition modules, eight simulated modules based on a SDM are used to evaluate all three correction procedures. The model parameters and modules are detailed in Tables S2 and S3. Equation (14) is the SDM formula, which describes the relationship between module current and voltage.

$$I = I_{ph} - I_0 \times \left[\exp\left(\frac{V + IR_s}{nN_s V_T}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \quad (14)$$

where N_s is the number of cells in PV modules; V_T is the thermal voltage; I_{ph} is the photocurrent, I_0 is the saturation current, and n is the diode factor. These parameters are detailed in Section C of Supporting Information S1.

To evaluate the universal applicability of the model, two real modules (1 TOPCon and 1 HJT module) are used. The information of these two modules is shown in Section D of Supporting

Information S1. The I - V curves of each test sample are displayed in Figure 1.

2.3 | Performance Evaluation Metrics

Equation (15) is used to quantify the discrepancy between the corrected and reference I - V curve:

$$X_{dev} = \frac{X_c - X_{ref}}{X_{ref}} \times 100\% \quad (15)$$

where X_{dev} is the percentage deviation in a parameter. Here, we use this for the differences in I_{SC} , V_{OC} , and P_{MAX} of the corrected I - V curves compared the values extracted from the reference curve; X_c and X_{ref} correspond to these parameters as extracted from the corrected and the reference I - V curves, respectively.

To evaluate and validate the procedures in terms of agreement in I_{SC} , V_{OC} , and P_{MAX} with their values from the reference I - V curve, we have employed the root mean squared error (RMSE) for I_{SC} , V_{OC} , and P_{MAX} . In previous research, this method has

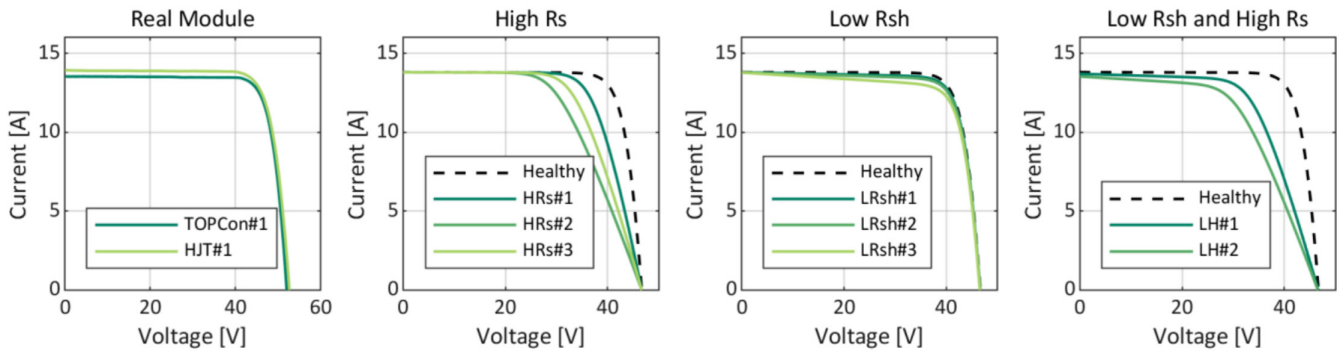


FIGURE 1 | Examples of I - V curves of real module and the simulated modules with healthy, high R_s , low R_{sh} , and both low R_{sh} and high R_s at STC.

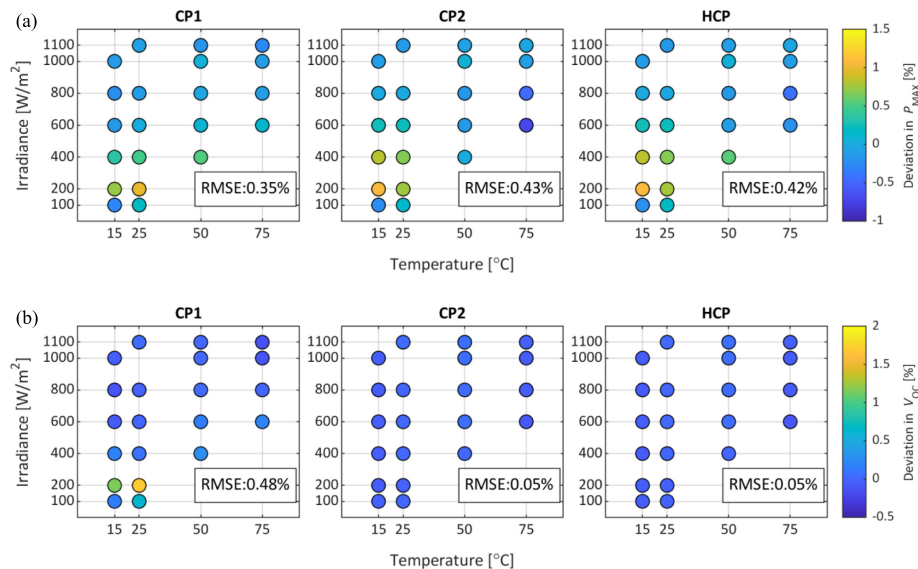


FIGURE 2 | (a) The deviation in P_{MAX} of translated I - V curves to STC using the three different CP for TOPCon#1; (b) the deviation in V_{OC} of translated I - V curves to STC using the three different CP for TOPCon#1.

been effectively utilized for I - V translation comparisons [11]. RMSE was determined by the following formulae:

$$\text{RMSE} = \sqrt{\frac{\sum_i^n (X_{dev})^2}{n}} \quad (16)$$

3 | Correction Performance

To assess the performance of each correction method across a wide range of irradiance (G) and module temperature (T_m) conditions, as well as to analyze the influence of G and T_m , all test samples were either simulated or measured under seven irradiance

levels (100, 200, 400, 600, 800, 1000, and 1100 W/m²) and four temperature levels (15°C, 25°C, 50°C, and 75°C). The selection of irradiances and temperatures corresponds to the complete irradiance-temperature matrix defined in IEC 61853-1 [19], resulting in 22 different reporting test conditions including STC. The I - V curves obtained under these environmental conditions were subsequently corrected to STC using CP1, CP2, and HCP methods.

3.1 | PV Module With Healthy Condition

As illustrated in Figure 2, CP1, CP2, and HCP exhibit comparable performance in correcting P_{MAX} for the real and healthy

TABLE 2 | RMSE of I_{SC} , V_{OC} , and P_{MAX} produced by CP1, CP2, and HCP. (For samples LH#1 and LH#2, results with $G < 400$ W/m² are excluded due to significantly distorted or invalid correction, as illustrated in Figure S2 in Section E).

Parameter	I_{SC}			V_{OC}			P_{MAX}		
Method	CP1	CP2	HCP	CP1	CP2	HCP	CP1	CP2	HCP
TOPCon#1	0.40%	0.35%	0.34%	0.48%	0.05%	0.05%	0.35%	0.43%	0.42%
HJT#1	0.23%	0.28%	0.25%	0.44%	0.07%	0.07%	0.46%	0.43%	0.42%
Healthy	0.02%	0.05%	0.02%	1.19%	0.02%	0.03%	0.01%	0.34%	0.45%
HRs#1	0.02%	0.05%	0.02%	1.02%	0.02%	0.02%	0.01%	0.34%	0.40%
HRs#2	0.03%	0.10%	0.02%	1.95%	0.02%	0.02%	0.02%	0.34%	0.39%
HRs#3	0.04%	0.15%	0.03%	2.92%	0.02%	0.02%	0.04%	0.32%	0.33%
LRsh#1	0.19%	0.56%	0.21%	1.55%	0.04%	0.04%	0.02%	4.99%	0.16%
LRsh#2	0.28%	0.92%	0.31%	1.86%	0.06%	0.06%	0.02%	7.70%	0.23%
LRsh#3	0.54%	2.48%	0.62%	3.48%	0.17%	0.16%	0.01%	15.41%	0.45%
LH#1	0.35%	0.45%	0.31%	0.69%	0.05%	0.05%	0.02%	1.15%	0.18%
LH#2	0.78%	1.03%	0.71%	1.39%	0.14%	0.14%	0.32%	1.74%	0.61%

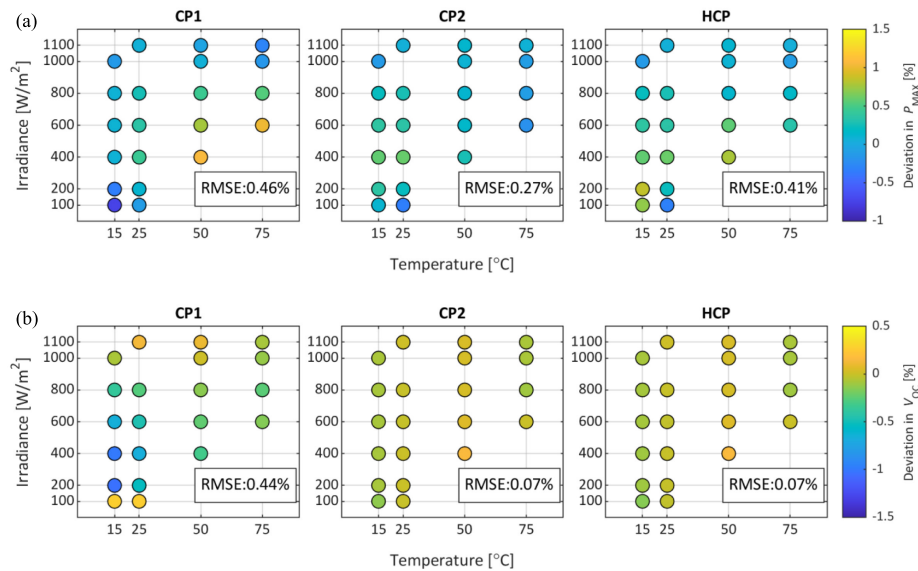


FIGURE 3 | (a) The deviation in P_{MAX} of translated I - V curves to STC using the three different CP for HJT#1; (b) the deviation in V_{OC} of translated I - V curves to STC using the three different CP for HJT#1.

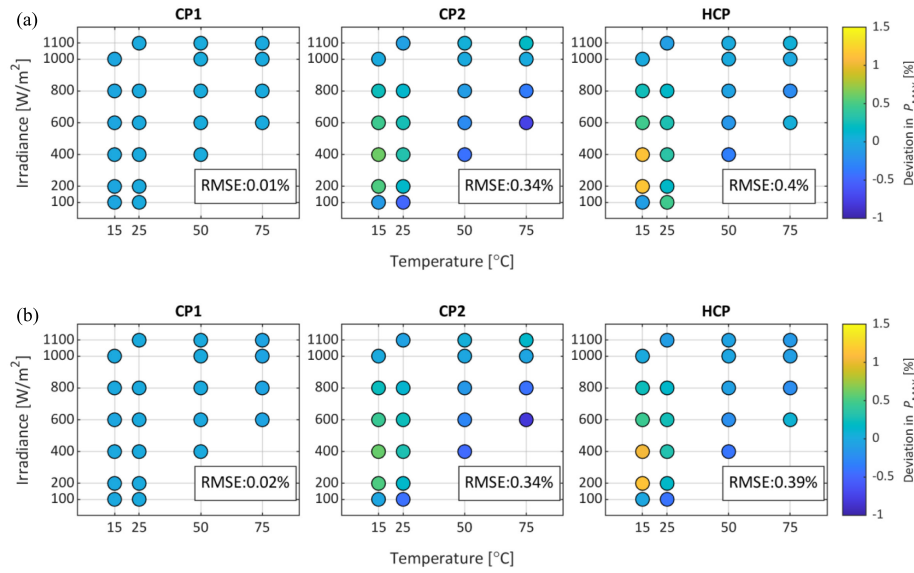


FIGURE 4 | (a) The deviation in P_{MAX} of translated I - V curves to STC using the three different CP for HRs#1; (b) the deviation in P_{MAX} of translated I - V curves to STC using the three different CP for HRs#2.

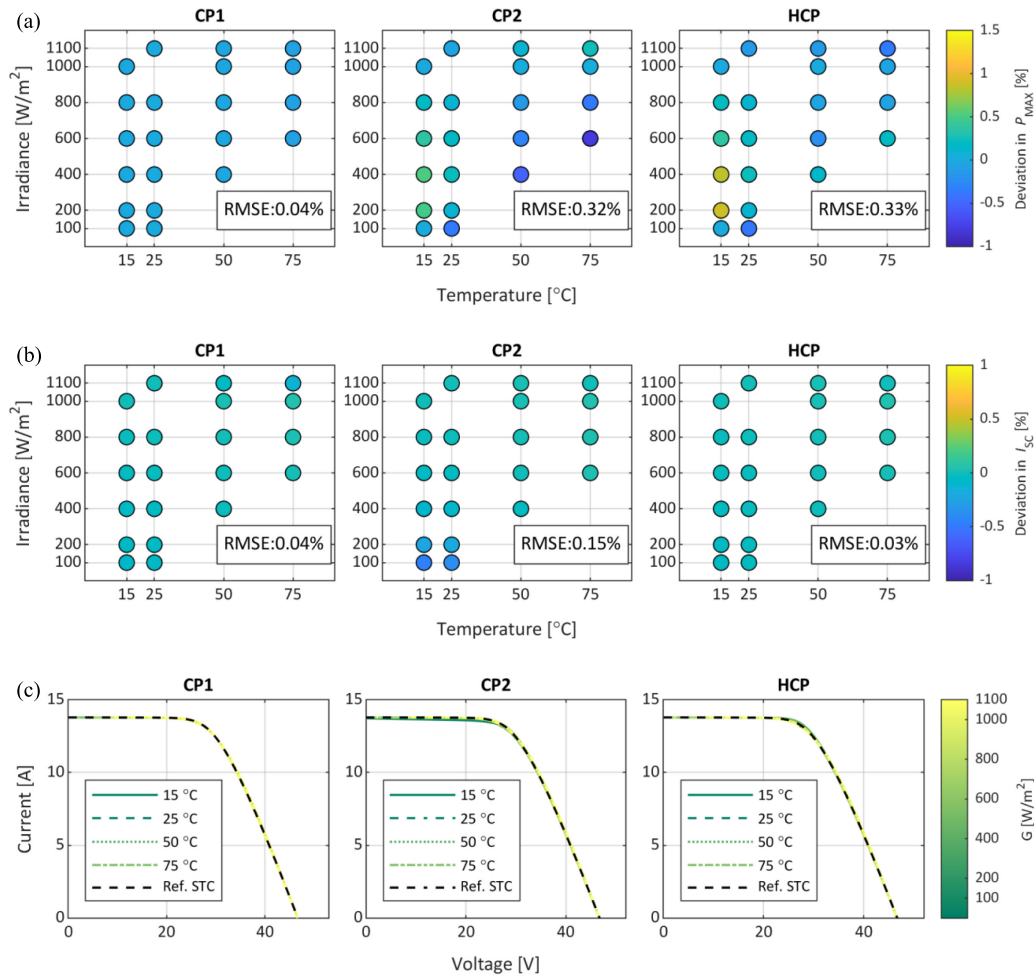


FIGURE 5 | (a) The deviation in P_{MAX} of translated I - V curves to STC using the three different CP for HRs#3; (b) the deviation in I_{SC} of translated I - V curves to STC using the three different CP for HRs#3; (c) corrected I - V curves by each CP under various G - T_m conditions to STC for HRs#3.

modules. Additionally, as shown in Table 2 and Figures 2b and 3b, HCP produces similar results to CP2 in correcting V_{OC} . Previous studies, such as the work by Xu et al. [11], have demonstrated that CP2 generally yields superior accuracy in V_{OC} correction. This is attributed to a nonlinear scaling of V_{OC} against logarithmic irradiance, and an irradiance-dependent TC of V_{OC} introduced in CP2 [6]. The results obtained from two real modules preliminarily validate the effectiveness of the proposed HCP strategy. These findings further demonstrate that HCP achieves reliable correction outcomes for both a healthy reference module and a real-world operational module, highlighting its potential as a robust alternative correction method.

3.2 | PV Module With High Series Resistance

As shown in Table 2 and Figures 4 and 5a, CP1 produces the most accurate P_{MAX} and I_{SC} , where the maximum RMSE in P_{MAX} and I_{SC} is 0.04% among three high- R_s modules. This is because CP1 more closely aligns with the SDM according to its formulation.

Figure 5c shows that all the I - V curves corrected by all three CPs under various G - T_m conditions align closely with the reference STC I - V curve. The RMSE of CP1 in I_{SC} is 2.92% for test sample HRs#3, while the HCP and CP2 produce 0.02% RMSE. CP1 fails to accurately translate the V_{OC} region under low-irradiance conditions ($\leq 200 \text{ W/m}^2$) as shown in Figure 5b, resulting in highly inaccurate V_{OC} values because of incomplete corrected I - V curves. This limitation is consistent with findings from previous research [11].

CP2 and HCP produce comparable results to CP1 in terms of P_{MAX} accuracy. For V_{OC} , HCP overcomes the limitations of CP1 and achieves the same results as those of CP2, as shown in Table 2. These findings demonstrate that high- R_s modules have minimal impact on the accuracy of I - V curve translation for the HCP method.

3.3 | PV Module With Low Shunt Resistance

CP1 remains the most accurate method for calculating P_{MAX} and I_{SC} . However, for low- R_{sh} modules, higher temperatures in the original conditions result in reduced accuracy in I_{SC} ,

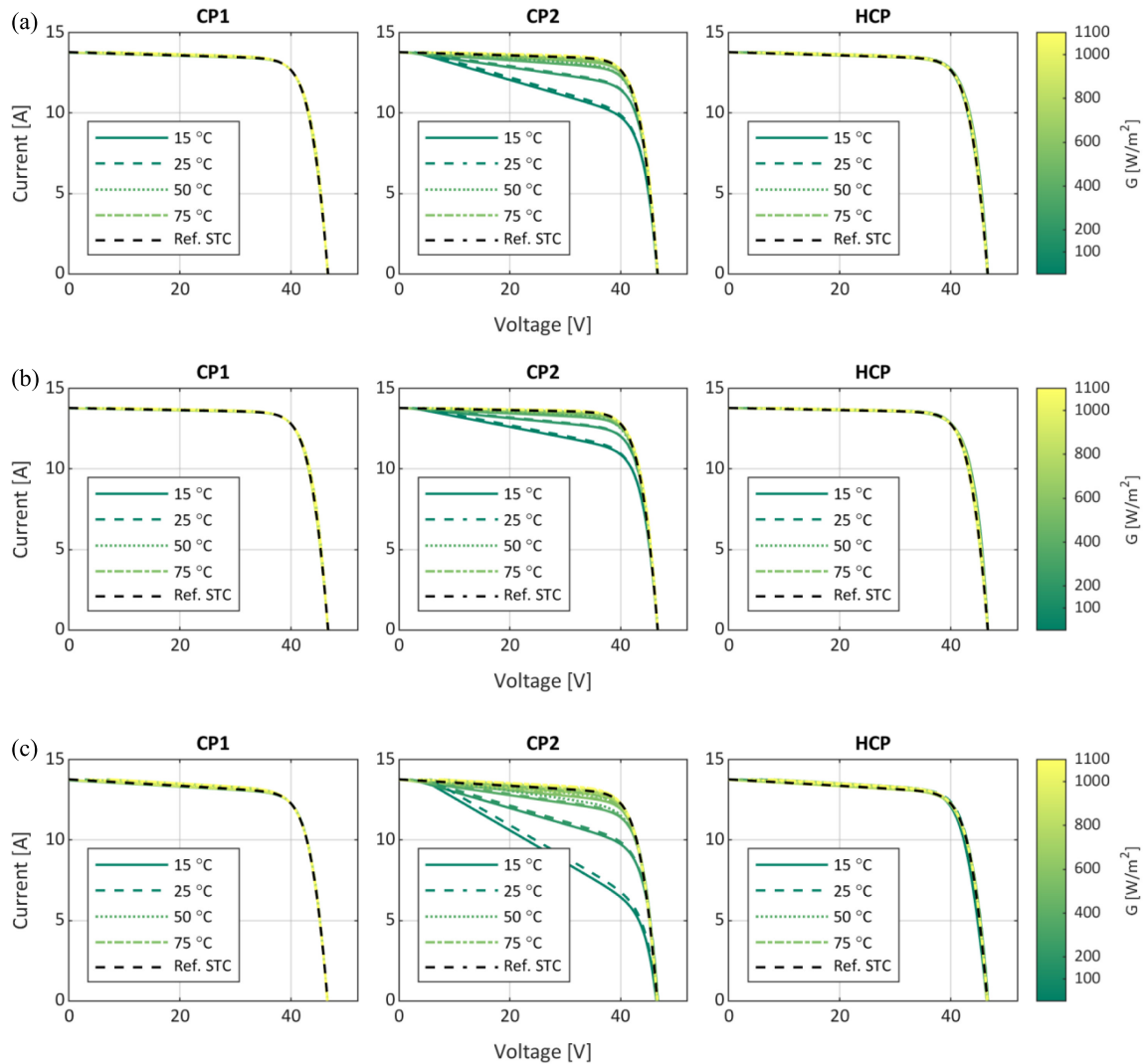


FIGURE 6 | Corrected I - V curves by each CP under various G - T_m conditions to STC for LRsh#1, LRsh#2, and LRsh#3, respectively.

which can be attributed to the mathematical formulation of Equation (2) [11].

Under high- R_s conditions, CP2 continues to deliver accurate results across all three electrical parameters (I_{SC} , V_{OC} , and P_{MAX}). However, for low- R_{sh} conditions, CP2 generates distorted $I-V$

curves, as shown in Figure 5, leading to significant deviations in P_{MAX} and impacting I_{SC} accuracy. The lower the irradiance in the original condition, the larger the deviation in P_{MAX} and I_{SC} calculations. The term $\frac{G_2}{G_1}$ in the Equation (3) plays a critical role in introducing these unreliable correction results of CP2 [11], as noted in Xu et al. Nonetheless, CP2 achieves accurate V_{OC}

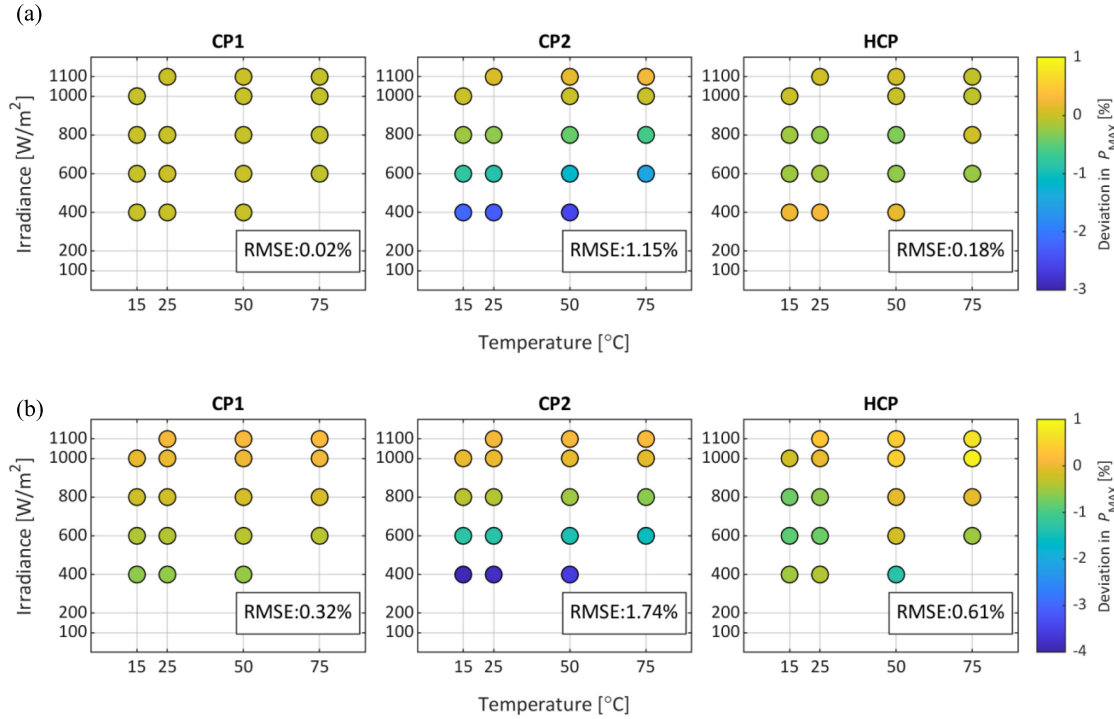


FIGURE 7 | (a) The deviation in P_{MAX} of translated $I-V$ curves to STC using the three different CP for LH#1; (b) the deviation in P_{MAX} of translated $I-V$ curves to STC using the three different CP for LH#2.

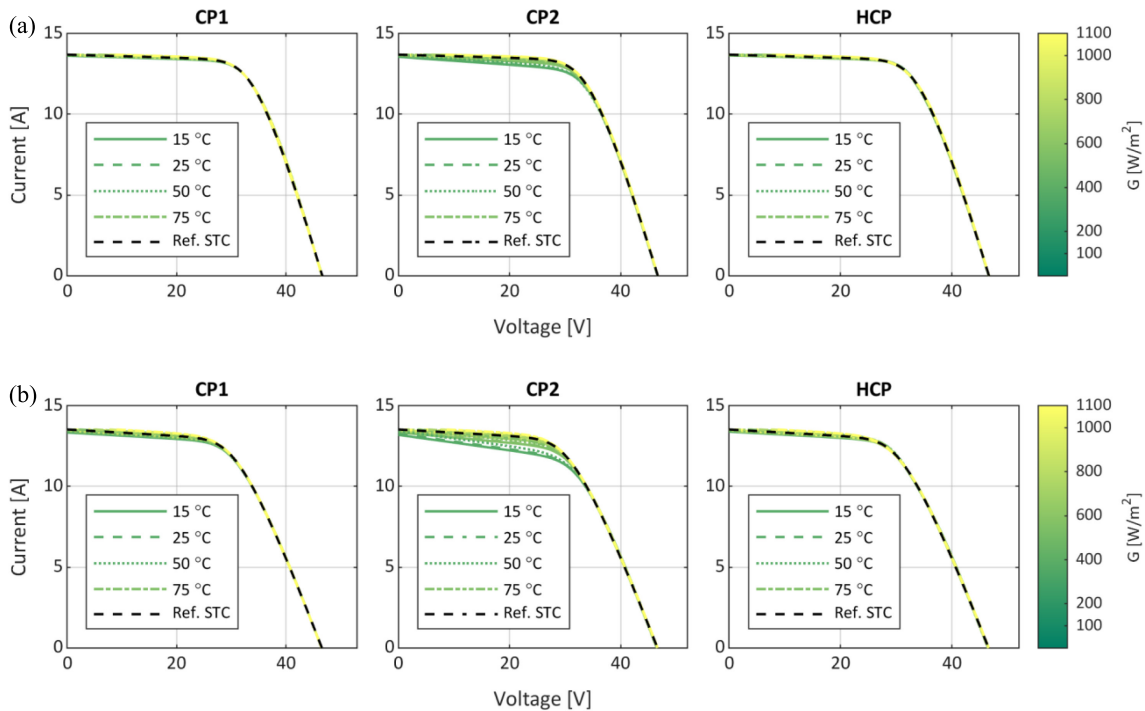


FIGURE 8 | Corrected $I-V$ curves by each CP under various $G-T_m$ conditions to STC for LH#1 and LH#2, respectively.

calculations, with the maximum deviation being only -0.08% across all low- R_{sh} test samples.

HCP demonstrates strong performance under low- R_{sh} conditions, with RMSE values for P_{MAX} of 0.16%, 0.23%, and 0.45% for Lsh#1, Lsh#2, and Lsh#3, respectively. The results indicate a trend where lower R_{sh} values correspond to reduced accuracy in P_{MAX} , with similar trends observed for I_{SC} and V_{OC} . As shown in Table 2, HCP generates I_{SC} results comparable to CP1 and V_{OC} results similar to CP2. These findings demonstrate that HCP effectively integrates the strengths of CP1 and CP2, with the I_{SC} region aligning more closely with CP1 and the V_{OC} region aligning more closely with CP2. Figure 5 further illustrates that all corrected I - V curves using HCP align closely, highlighting its accuracy across a broad range of G and T_m conditions (Figure 6).

3.4 | PV Module With High Series Resistance and Low Shunt Resistance

CP1 achieves the lowest RMSE in P_{MAX} among the three correction procedures for all simulated modules, with RMSE values below 0.05%—except for sample LH#2. For LH#2, CP1 produces an RMSE of 0.32% in P_{MAX} . As demonstrated in Section 3.2, high R_s on its own has minimal influence on correction accuracy. However, the results of Figure 7 and Table 2 suggest that the combination of low R_{sh} and high R_s significantly affects correction accuracy more than either factor individually.

As shown in Figure 8, CP2 fails to provide accurate I - V curve translation due to the influence of low R_{sh} , whereas HCP achieves better alignment with all the corrected I - V curves. HCP achieves comparable P_{MAX} results with significantly lower RMSE than CP2 in the other irradiance levels, as shown in Table 2. Furthermore, HCP delivers highly accurate V_{OC} calculations, similar to CP2. For I_{SC} , HCP's accuracy is influenced by the CP1-based approach, which is less accurate under challenging conditions such as high R_s or low R_{sh} . However, the I_{SC} results remain comparable to those of CP1.

4 | Conclusions

This study presents a comprehensive evaluation of three I - V curve correction methods—CP1, CP2, and the newly proposed HCP—under a wide range of irradiance and temperature conditions, including real TOPCon and HJT modules and simulated modules with varying R_{sh} and R_s .

CP1 consistently produces the accurate P_{MAX} calculations across all simulated conditions, with RMSE values below 0.50% for all modules. Although CP1 exhibits limitations in I_{SC} calculations for low- R_{sh} modules, particularly under high-temperature conditions, it still achieves accurate I_{SC} estimations for these modules, with an RMSE of less than 0.54%. This reduced accuracy is attributed to inherent shortcomings in CP1's mathematical formulation. Additionally, a major limitation of CP1 is its inability to produce complete I - V curves when translating from low- to high-irradiance levels, a drawback that has also been highlighted in previous research [11].

CP2 achieves reliable performance for V_{OC} correction across all modules, even in challenging conditions, with a maximum deviation of only -0.27% . However, it struggles with accurate I - V curve translations for low- R_{sh} modules, producing significant deviations in P_{MAX} and I_{SC} , particularly under low-irradiance scenarios.

The proposed HCP method successfully integrates the strengths of CP1 and CP2, achieving comparable I_{SC} results to CP1 and V_{OC} results similar to CP2. As shown in Table 2, HCP achieves identical RMSE values for V_{OC} across all test samples, while the difference in RMSE for I_{SC} between CP1 and HCP remains below 0.1% for all samples. HCP demonstrates strong performance across a broad range of G - T_m conditions, showing improved accuracy for low- R_{sh} and high- R_s modules compared to CP1 and CP2. However, it is important to note that for modules with both low R_{sh} and high R_s , HCP introduces minor distortions in I - V curves at low-irradiance levels ($G \leq 200 \text{ W/m}^2$) to high-irradiance levels. Despite this limitation, HCP achieves lower RMSE in P_{MAX} than CP2 and maintains high accuracy in V_{OC} corrections under other G - T_m conditions.

Overall, the HCP method offers a robust alternative for I - V curve correction for TOPCon and HJT c-Si PV technology modules, combining the advantages of CP1 and CP2 while addressing their respective limitations. This approach is also expected to achieve accurate results for other c-Si technologies, such as PERC and Al-BSF, as both CP1 and CP2 have already been evaluated for these technologies [6, 11]. The results highlight the importance of considering both R_s and R_{sh} effects in low G and high T_m environments to ensure accurate correction. The results indicate that the HCP method has the potential to deliver precise I - V curve corrections for PID-affected modules, including both fill factor-affected (PID-s) and I_{SC} -affected (PID-p) cases. Future work may focus on further optimizing the HCP approach to minimize distortions under extreme conditions and on extending its validation to additional real-world PV datasets with the final aim of enhancing its applicability in real scenarios.

Author Contributions

Wenhao Xu conceived the idea presented and developed the formulas. Wenhao Xu also carried out the experiments and performed numerical calculations. Christos Monokroussos encouraged Wenhao Xu to investigate this research and supervised the findings of this work. Wenhao Xu took the lead in writing the manuscript. All authors discussed the results and contributed to the final manuscript.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author, Wenhao Xu (frank.xu@tuv.com), upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** The diagram of $Y1(V)$ and $Y2(V)$ with $V_k = 40.2$ and $k = 0.45$. Both curves range from 0 to 1, exhibiting different trends to facilitate the integration. **Figure S2:** Example of distorted I-V curve produced by HCP for test sample LH#2. **Table S1:** The value of i used in the paper. **Table S2:** Single-diode model parameters for the simulated models. **Table S3:** The information of two real sample modules.